SignBridge AI: Real-Time Deep Learning ASL Recognition

Capstone Project Proposal — PGA 3.0 (Track 2: Machine Learning & Deep Learning)

1. Executive Summary

Millions of Deaf and hard-of-hearing individuals communicate primarily in American Sign Language (ASL), yet most everyday settings, workplaces, and public services lack real-time interpretation support. SignBridge AI is a deep learning system that recognizes ASL fingerspelling (the 26-letter alphabet) from static images and a curated set of common word-level signs from short video clips, translating recognized signs into on-screen text through a webcam-based Streamlit application.

2. Problem Statement

Background: ASL is a complete, distinct language with its own grammar and is the primary language for many Deaf individuals. Real-time human interpretation tools remain expensive or unavailable on demand. Advances in MediaPipe hand-landmark tracking and sequence-based deep learning make lightweight, accessible sign-recognition tools highly feasible.

Objective: To build and deploy a deep learning system that (a) classifies static ASL alphabet gestures from images with high accuracy, and (b) recognizes a curated subset of 25-30 common word-level ASL signs from video clips by modeling motion of hand landmarks over time, presented through a real-time webcam Streamlit demo.

Scope: The project covers the 26-letter ASL alphabet for static recognition, and 25-30 high-frequency word-level signs (e.g., 'hello', 'thankyou', 'yes', 'no', 'please', 'help') for dynamic video recognition. Full-sentence translation with facial grammar is explicitly out of scope.

3. Data Sources

Primary Data: N/A

Secondary Data - Sign Language MNIST (Kaggle): 28x28 grayscale hand-gesture CSV dataset covering 24 static ASL alphabet letters. Pre-cleaned, public Kaggle dataset (`datamunge/sign-language-mnist`).

Secondary Data - WLASL Dataset Mirror (Kaggle): Pre-processed video mirror of Word-Level ASL dataset (`risangbaskoro/wlasl-processed` on Kaggle), filtered to 25-30 target word glosses.

4. Methodology

Data Collection: Downloaded directly from Kaggle public research repositories (`datamunge/sign-language-mnist` and `risangbaskoro/wlasl-processed`).

Data Preparation: For Sign Language MNIST: normalize 28x28 pixel vectors and apply rotation/scaling augmentation. For WLASL video clips: extract 21 hand keypoint (x, y, z) landmarks per frame using Google MediaPipe, converting raw video into lightweight normalized 63-dimensional coordinate sequences.

Analysis & Modeling Techniques: ['● Baseline Classical ML: Random Forest and Support Vector Machines (SVM) trained on hand-crafted landmark features.', '● Static Alphabet Model: A 2D Convolutional Neural Network (CNN) trained on 28x28 ASL alphabet pixel arrays.', '● Dynamic Sequence Model: An LSTM / GRU recurrent neural network trained on MediaPipe landmark sequence vectors to capture movement patterns over time.', '● Evaluation Metrics:

Accuracy, Precision, Recall, Macro F1-score, and Confusion Matrix.']

Tools: Python, Google MediaPipe, TensorFlow / PyTorch, scikit-learn, OpenCV, and Streamlit for real-time webcam app deployment.

5. Expected Outcomes

- A trained CNN classifier for the static 26-letter ASL alphabet achieving >95% F1-score.
- A trained LSTM/GRU sequence model recognizing 25-30 dynamic word-level ASL signs from landmark vectors.
- Comparative performance benchmarks comparing classical ML on static landmarks vs. deep recurrent sequence models.
- A deployed interactive Streamlit application featuring live webcam sign recognition, landmark skeleton rendering, and real-time text output.

6. Risks and Challenges

Signer & Lighting Variation: Different hand sizes, lighting, and camera angles. Mitigation: MediaPipe landmark coordinate normalization is invariant to skin tone, background, and lighting.

Real-time Demo Latency: Raw video processing causing frame lag. Mitigation: Processing lightweight 63-numerical coordinate arrays instead of raw video pixels ensures <30ms real-time inference.

7. Conclusion

SignBridge AI combines high visual appeal, immediate real-time interactivity, and strong technical depth (CNN + LSTM + MediaPipe) while remaining fast to execute on clean public Kaggle datasets.